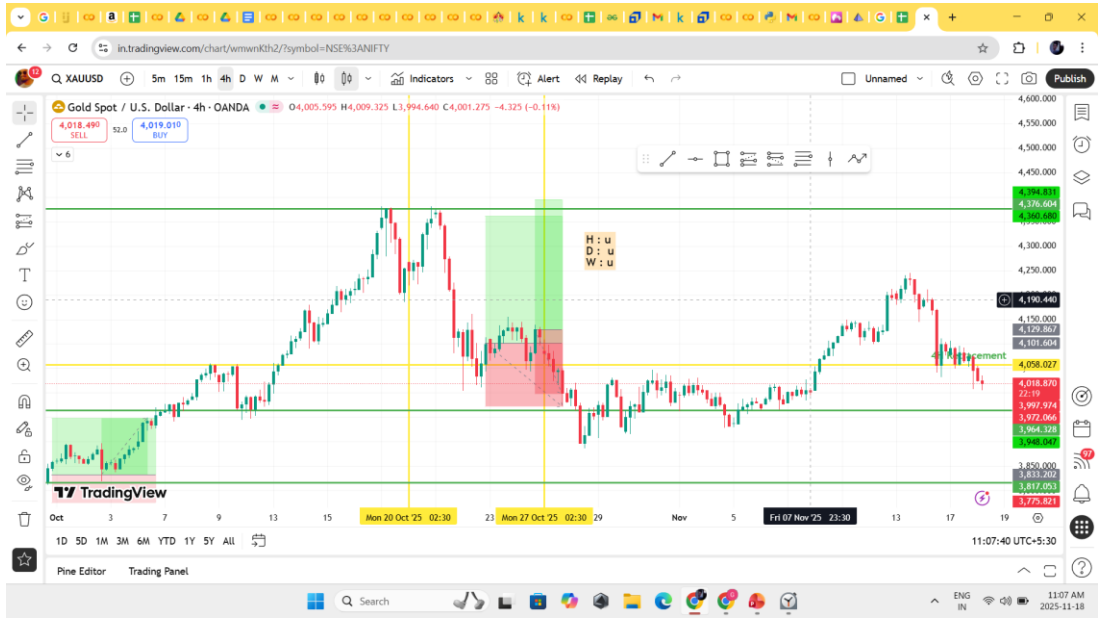
Optimizing Profitable Trades Of Forex Trading

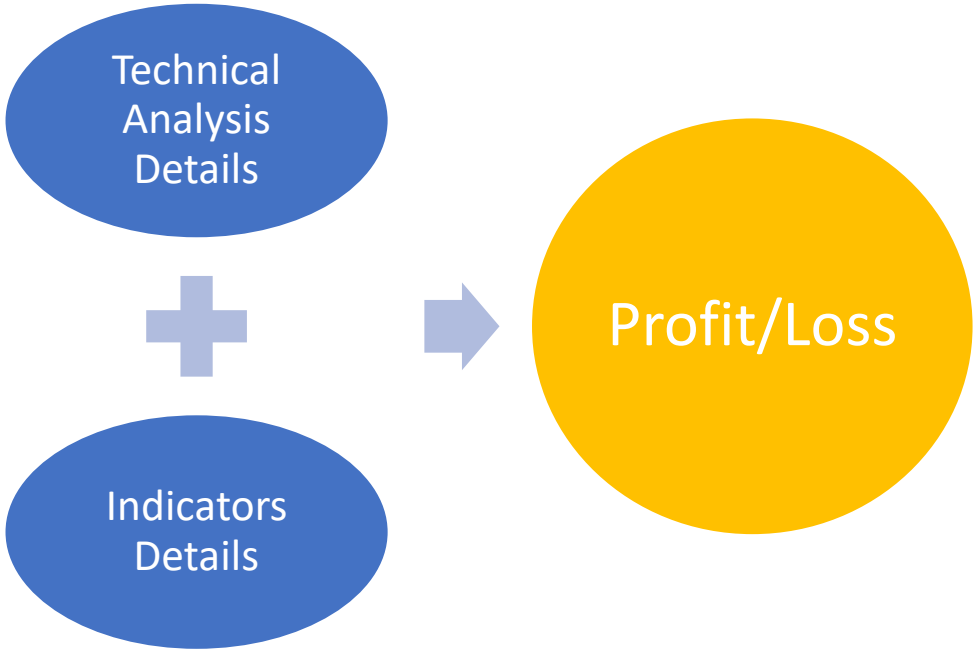
Gone Ramu (25RA06010)

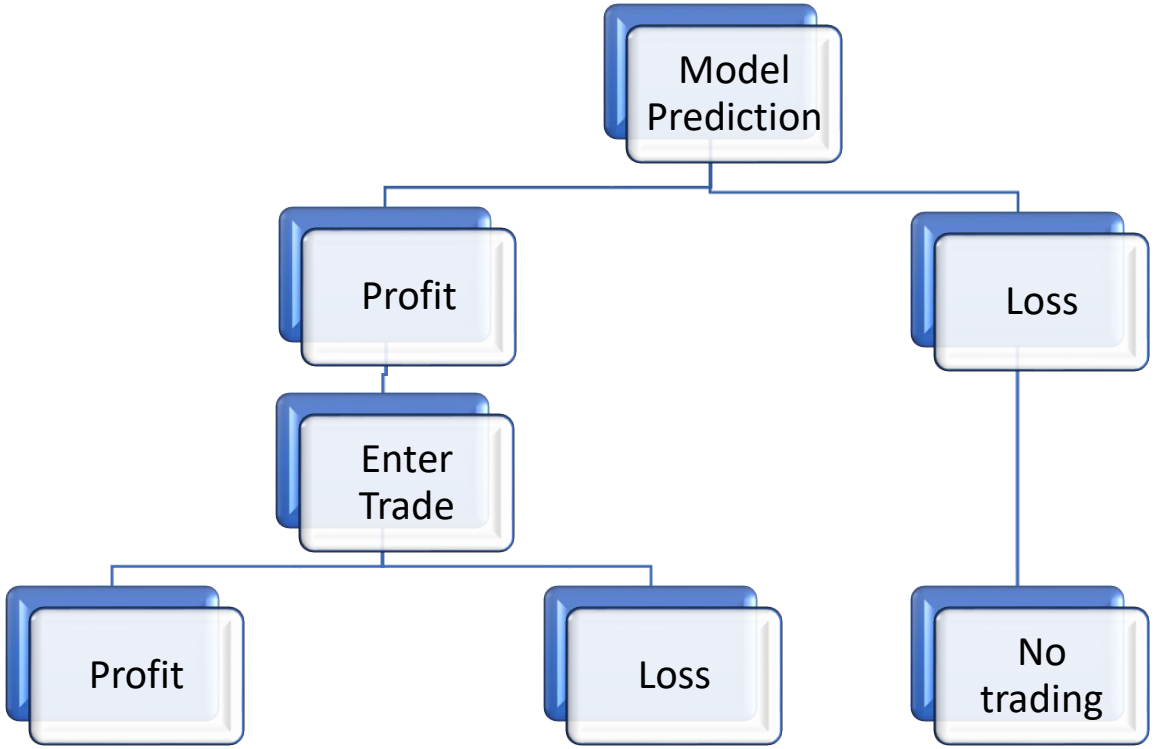
Problem Statement

- Given the technical analysis details and indicator signals of a trade entry, the goal is to build a machine learning model that predicts whether the trade will end in profit or loss.
- This helps traders enter only high-probability trades and avoid low-probability ones.
- I created my own dataset consisting of real trade entries recorded only near key support and resistance zones.



Problem Statement





Dataset Description

- Created own dataset by recording real forex trade entries manually.
- The dataset contains trade setups from January 2025 to present
  - AUDUSD, EURUSD, GBPUSD, NZDUSD, USDCAD, USDCHF,USDJPY , GOLD, BITCOIN
- Each row represents one complete trade with entry details, indicator signals and final outcome.
- **Target variable :**
  - Profit/SL
- **Feature categories :**
  - **Technical Analysis features:**
    - Trade details, Market structure, Candlestick information, etc.
  - **Technical indicators :**
    - RSI, MA, BB, Par, etc.

Dataset Preprocessing

Drop null values

- Dropping rows, columns

EDA Analysis

- Category, Numerical Cols
- Unique values & their count
- Kde plots of numerical cols
- Observed Target Variable has 55 – 45% distribution

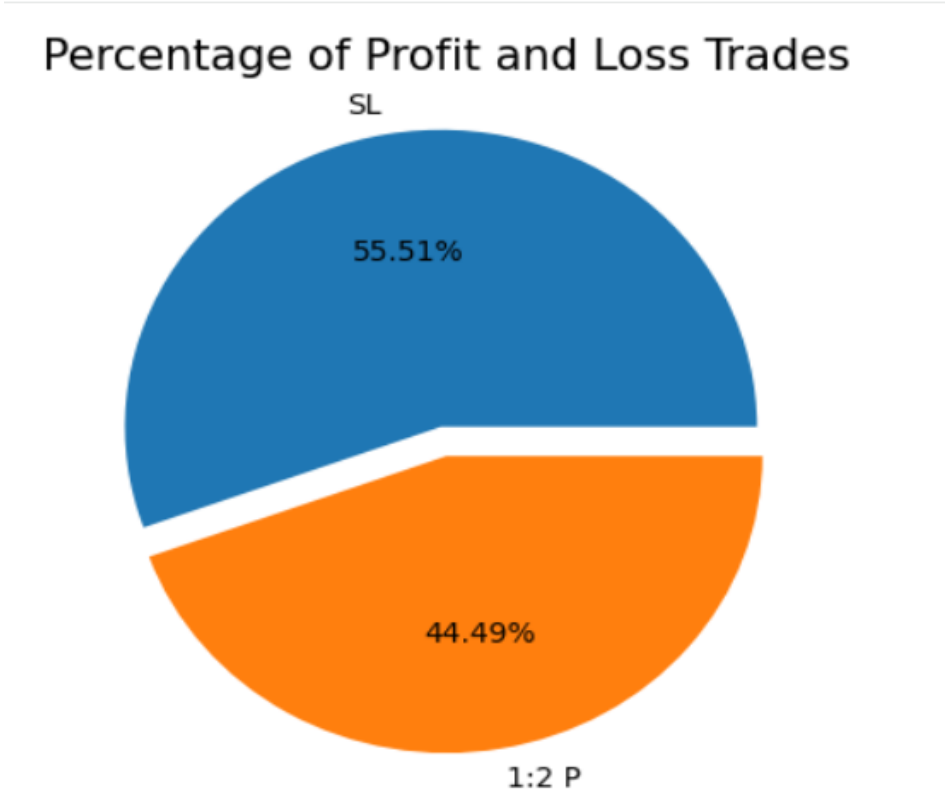
Data Preprocessing

- Target Encoding
- Standard Scaling

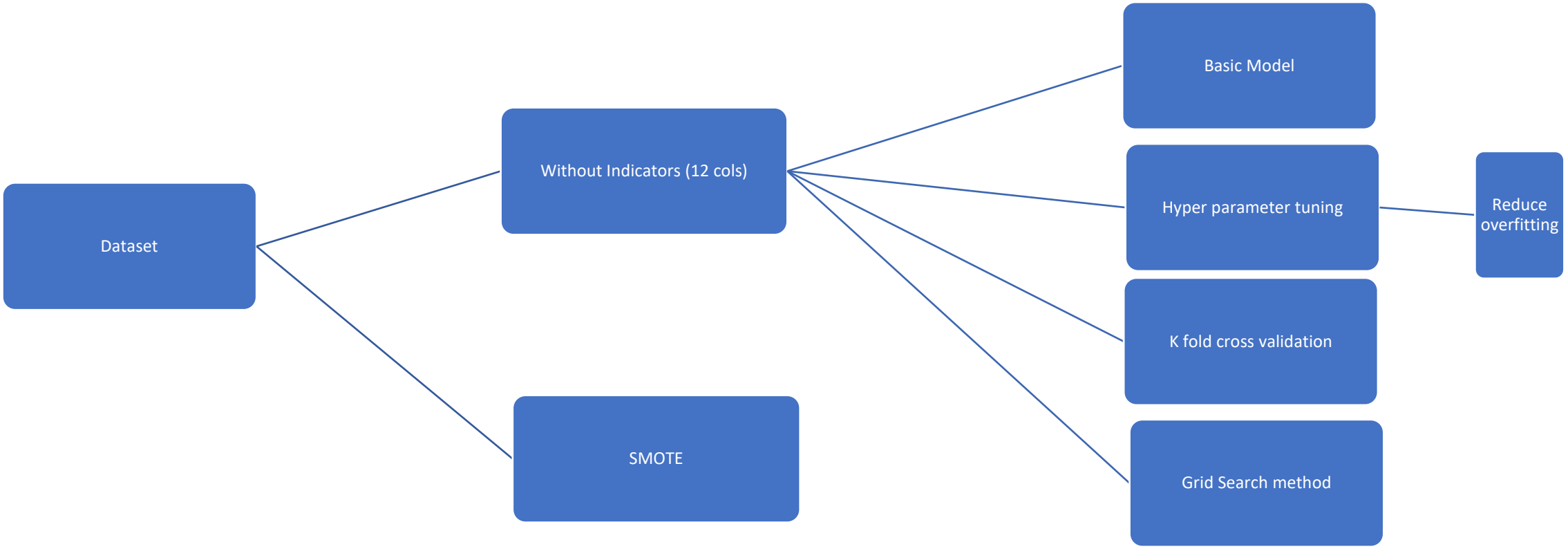
Train,Test,Split

- Without indicators (12 cols)

Class Distribution



Models Training Process



Logistic Regression

Random Forest

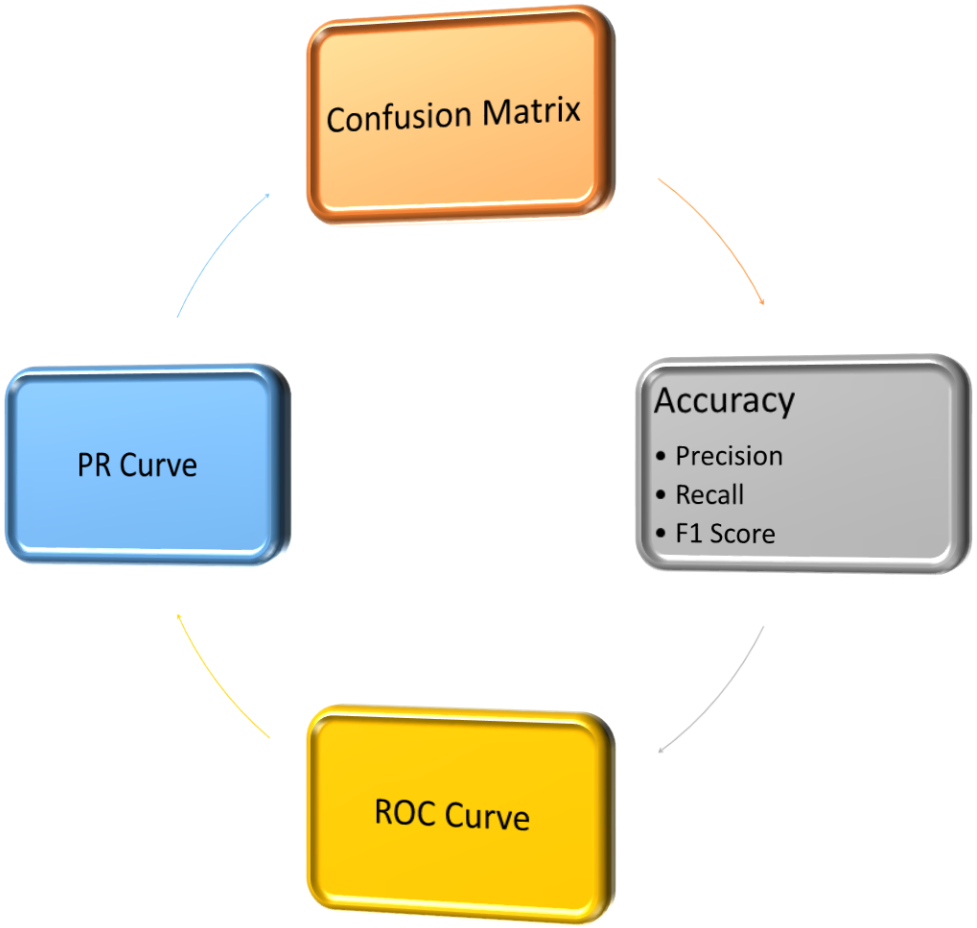
KNN

Decision Tree

Extra Models

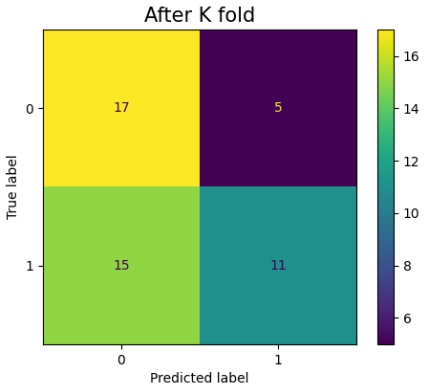
- XGB, LGBM
- CatBoost, SVC

Metrics Observation

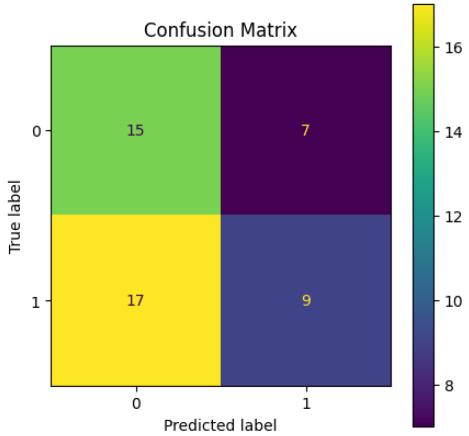


ML Models & Their Accuracy

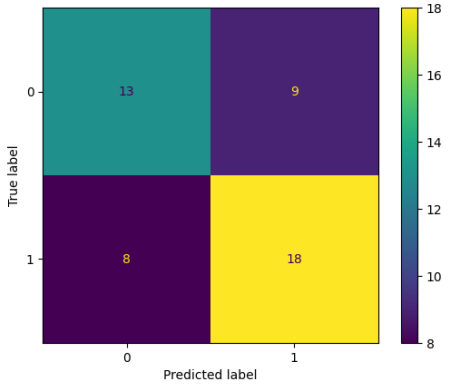
Model Name	Without indicators (12 cols)	SMOTE (Without indicators) (12 cols)
Logistic Regression	0.58	0.60
RandomForest	0.50	0.64
KNN	0.65	0.70
Decision Tree	0.60	0.62
XGB	-	0.70
CatBoost	-	0.62
LGBM	-	0.62
SVC	-	0.60
NN		0.49



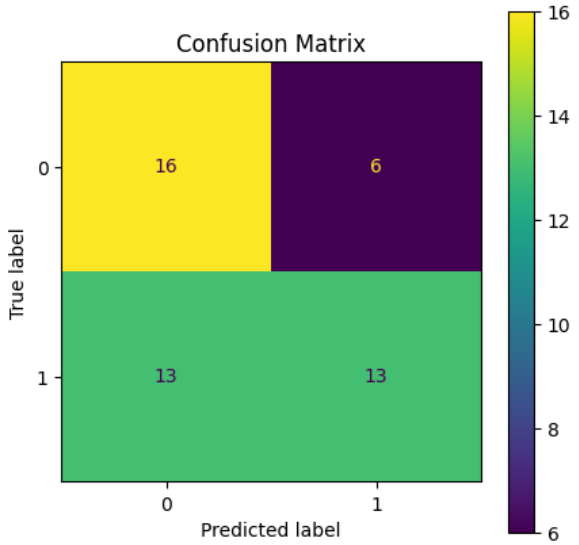
→ **Logistic Regression**



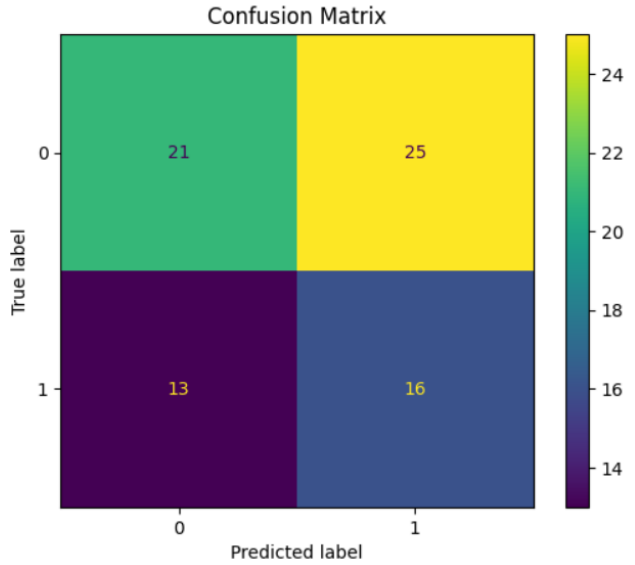
→ **Random Forest**



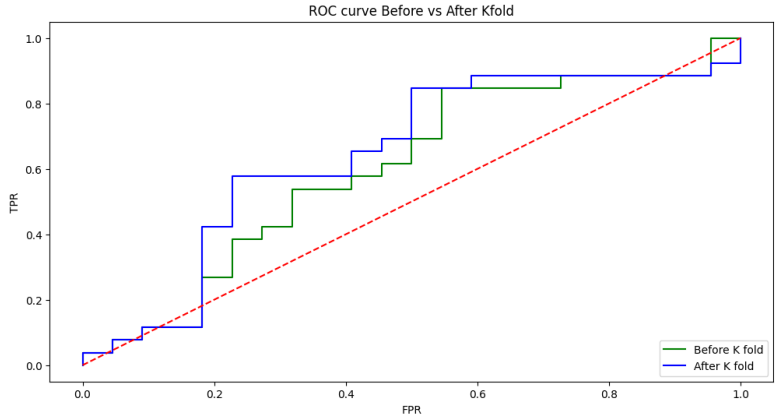
→ **KNN model**



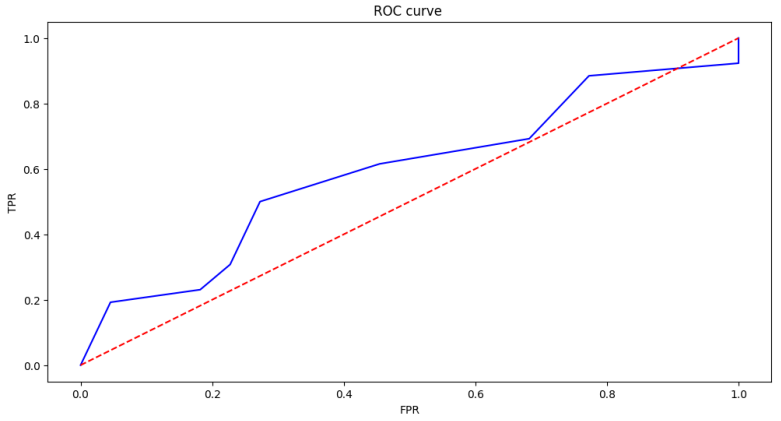
Decision Tree model



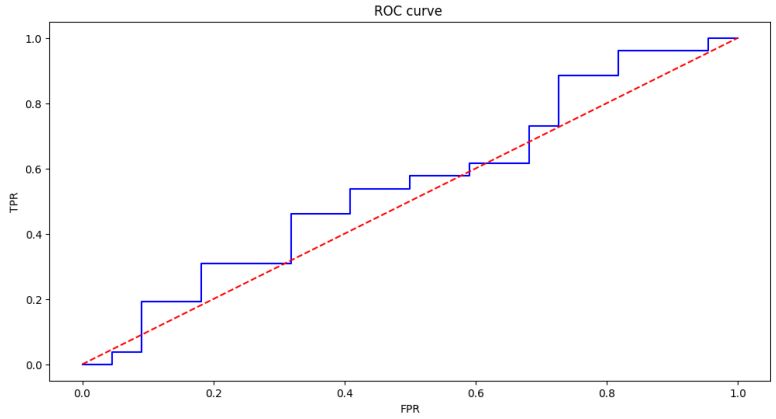
NN model



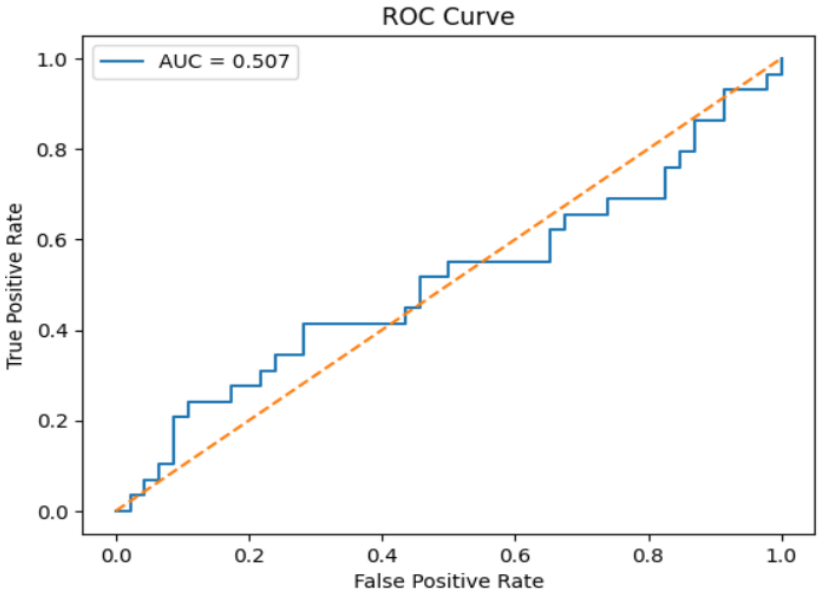
Logistic Regression



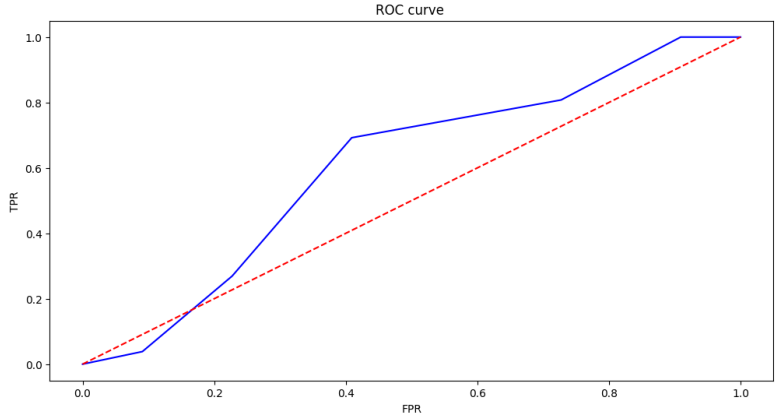
Decision Tree model



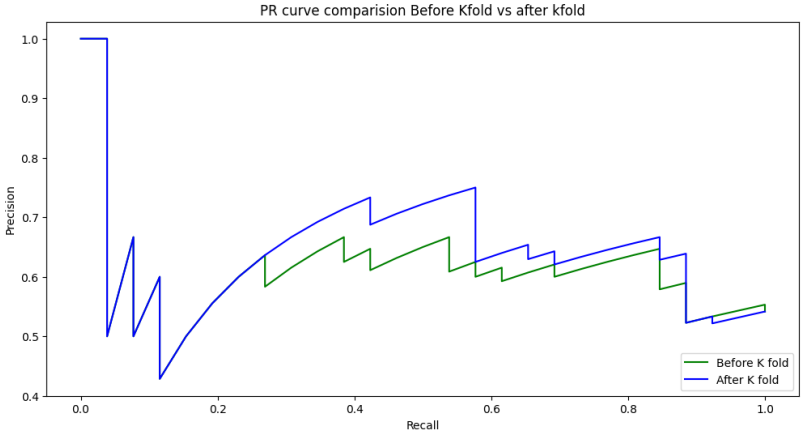
Random Forest



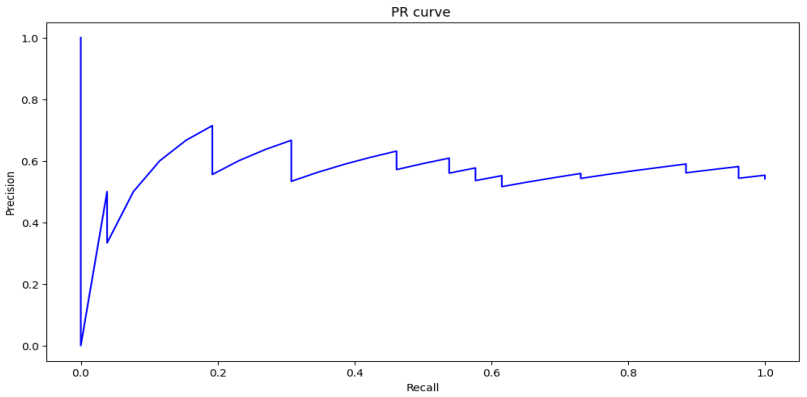
NN model



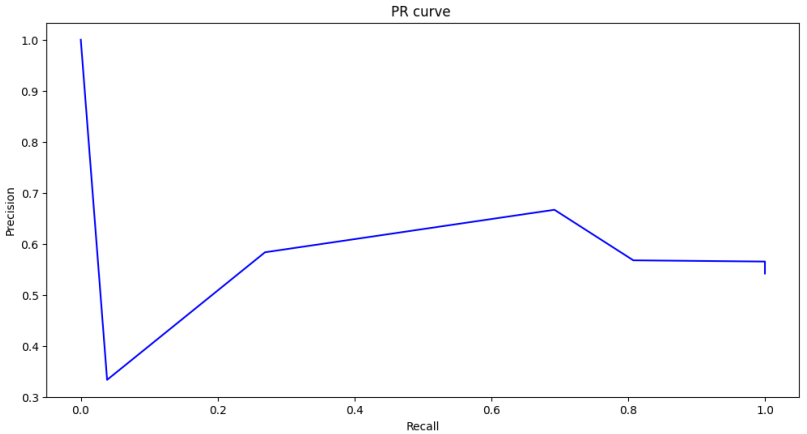
KNN model



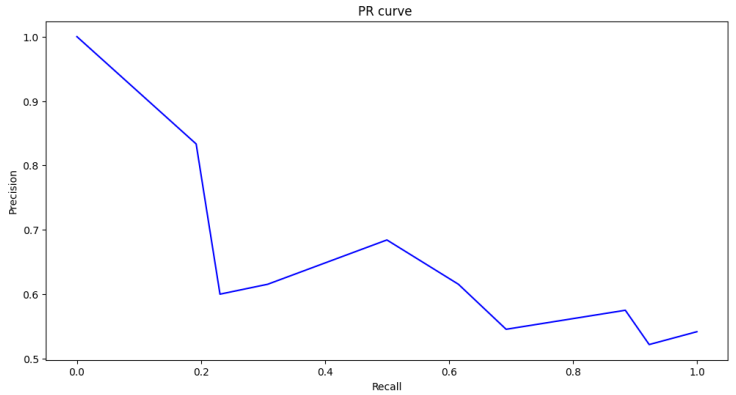
Logistic Regression



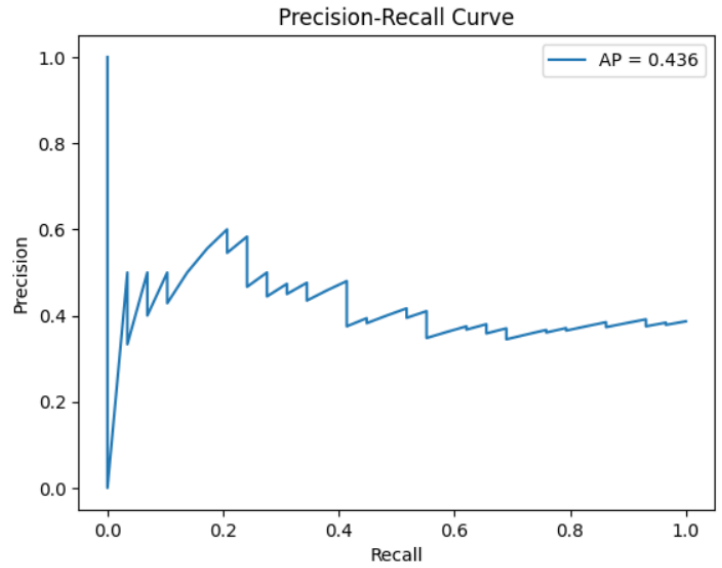
Random Forest



KNN model

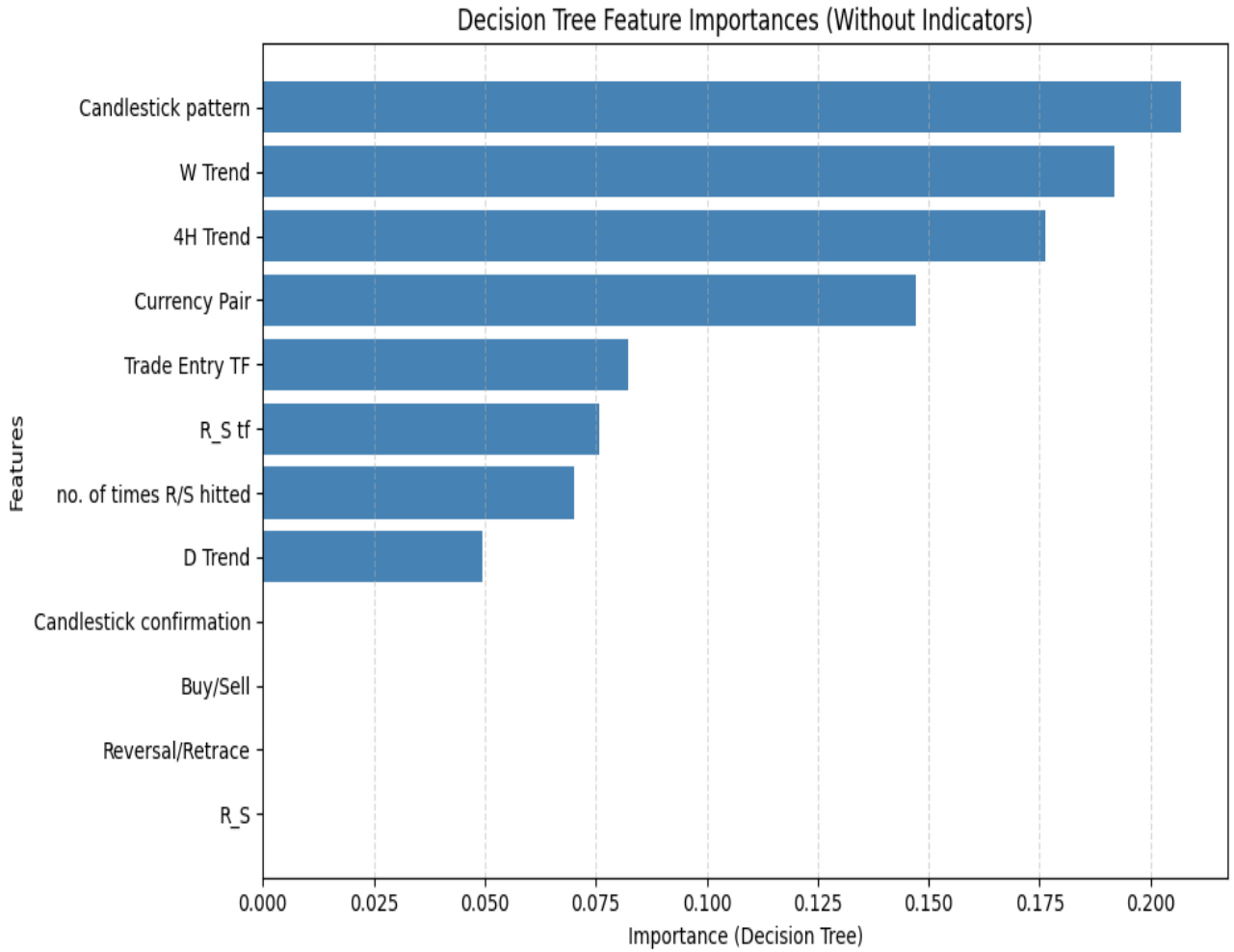


Decision Tree model



NN model

Feature Importance



ML Models and Accuracy

- Without indicators : KNN performed (65%) well than other models.
- With indicators : KNN (58%)showed better performance.
- SMOTE without indicators: KNN and XGBoost achieved highest accuracy (0.70).
- SMOTE with indicators: KNN achieved higher accuracy overall (65%).
- Given dataset is not linearly separable.
- KNN performed better as it learn non-linear behaviour.